
Chapter 8: Playing with Constructions

Worksheet 8: Constructing Quadrilaterals

Learning Objective: Students will construct quadrilaterals given different combinations of measurements (sides and angles, diagonals, etc.); understand conditions for unique construction; and apply constructions to solve geometric problems.

Worked Example

To construct a parallelogram $ABCD$ with $AB = 6$ cm, $BC = 4$ cm, $\angle B = 75^\circ$:

1. Draw $AB = 6$ cm.
2. At B , construct $\angle ABX = 75^\circ$.
3. Mark C on BX such that $BC = 4$ cm.
4. From A , draw $AY \parallel BC$ and from C , draw $CZ \parallel AB$.
5. Their intersection is D . Join AD and CD .

Questions 1–5: Easy / Recall

- Q1.** How many measurements are needed to uniquely construct a triangle?
- Q2.** How many measurements are needed to uniquely construct a quadrilateral?
- Q3.** Name three special conditions that allow a quadrilateral to be uniquely constructed with fewer measurements.
- Q4.** Construct a square of side 5 cm. (Teacher Verification Required)

- Q5.** Construct a rectangle with length 7 cm and breadth 4 cm. (Teacher Verification Required)

Questions 6–10: Basic Application

- Q6.** Construct a parallelogram $ABCD$ with $AB = 5$ cm, $BC = 3.5$ cm, $\angle A = 70^\circ$. (Teacher Verification Required)

- Q7.** Construct a rhombus with side 5 cm and one angle 60° . Find its diagonals by measurement. (Teacher Verification Required)

- Q8.** Construct a quadrilateral $ABCD$ where $AB = 4$ cm, $BC = 5$ cm, $CD = 6$ cm, $DA = 4.5$ cm, $\angle B = 90^\circ$. (Teacher Verification Required)

- Q9.** Construct a quadrilateral when its four sides and one diagonal are given: $AB = 4$, $BC = 6$, $CD = 5$, $DA = 5$, $AC = 7$ cm. (Teacher Verification Required)

- Q10.** Construct a trapezium $ABCD$ where $AB \parallel CD$, $AB = 7$ cm, $CD = 4$ cm, $AD = 3.5$ cm, $\angle A = 75^\circ$. (Teacher Verification Required)

Questions 11–15: Practice and Skill Building

- Q11.** Can a quadrilateral be constructed with sides 3, 4, 5, 6 and a diagonal of 10 cm? Explain why or why not.
- Q12.** Construct a quadrilateral $PQRS$ with $PQ = 5$ cm, $QR = 6$ cm, $RS = 7$ cm, $SP = 5$ cm, and $PR = 8$ cm. (Teacher Verification Required)
- Q13.** Describe the minimum information needed to uniquely construct a parallelogram. Construct one with $AB = 6$ cm and diagonal $AC = 8$ cm and $BD = 6$ cm. (Teacher Verification Required)
- Q14.** Construct a kite $ABCD$ where $AB = AD = 4$ cm and $CB = CD = 6$ cm. (Teacher Verification Required)

- Q15.** Construct a regular hexagon of side 4 cm using a circle. (Teacher Verification Required)

Questions 16–18: Higher Order Thinking

- Q16.** Is it possible to construct a quadrilateral with angles 80° , 70° , 110° , and 100° ? Verify and attempt construction. (Teacher Verification Required)
- Q17.** Why can't a quadrilateral be constructed with sides 2, 3, 4, and 12 cm? (Hint: use the quadrilateral inequality – sum of any three sides must be greater than the fourth.)
- Q18.** Construct a quadrilateral with two diagonals 6 cm and 8 cm intersecting at right angles, with the diagonals bisecting each other. What special quadrilateral is this? (Teacher Verification Required)

Questions 19–20: Real-Life Applications

- Q19.** An architect must design a room (on paper, to scale $1\text{ cm} = 1\text{ m}$) that is $5\text{ m} \times 4\text{ m}$. Construct the scaled rectangle and mark the positions of 2 doors of width 1 m each (one on the long wall, one on the short wall). (Teacher Verification Required)

- Q20.** A land plot is a quadrilateral $ABCD$ with $AB = 8$ m, $BC = 6$ m, $CD = 7$ m, $DA = 5$ m, and diagonal $BD = 9$ m. Construct it to scale (1 cm = 1 m). Find area by dividing into two triangles and using the construction. (Teacher Verification Required)

Reflection Question: What is the minimum number of measurements to uniquely construct any quadrilateral? Can you always construct a unique quadrilateral given 4 sides only? Why?